

# TENNIS MATRIX — FULL VERIFICATION AUDIT

## UPDATED WITH NEW VERIFIED WIN/LOSS LEARNING + CURRENT CALIBRATION RULES

## NEW POST-RESULT LEARNING / 120-MATCH CALIBRATION ENFORCEMENT

These rules are additive and override any earlier rule that would allow a candidate to bypass them.

### 1. UPDATED CALIBRATION POSITION

The current verified calibration/backtest must run AFTER the Verification Audit, Disagreement/Trap Audit, Dangerous-Underdog screen, stress tests, and removal case, but BEFORE the final color/output. Recompute from the newest verified sample every time it changes. Never reuse stale bucket rates.

### 2. NO-EDGE / LOW-PROBABILITY FLOOR

- Headline or independently reconstructed favorite probability  $\leq 55\%$ : automatic PASS/RED; never Green.
- 56-64%: cannot be Green unless the favorite defeats every major underdog pathway with at least two independent high-reliability evidence families and no unresolved critical contradiction.
- A Red/PASS classification means "selection is not reliable enough"; it does NOT automatically mean flip to the opponent.

### 3. INDEPENDENT FLIP STANDARD

Never FLIP solely because of one contradictory family, recent form alone, opponent-adjustment alone, H2H alone, ranking alone, or generic external narrative.

A FLIP requires:

- (a) at least 2 genuinely independent high-reliability underdog evidence families;
- (b) at least 1 independent external confirmation;
- (c) the underdog survives component-removal and conservative stress testing;
- (d) the favorite fails the strongest evidence-based defense.

If these are not met, DOWNGRADE/PASS rather than FLIP.

### 4. DANGEROUS UNDERDOG CEILING OVERRIDE

For every underdog, explicitly search for a recent ceiling event:

- win over Top 10/20/50 or clearly elite/high-level opponent;
- repeated competitive performances against much stronger opposition;
- rapid ranking/Elo acceleration;
- qualifying-to-main-draw or level-jump surge;
- major same-surface breakthrough;
- recent hold/break/serve/return performance materially above long-run rating;
- comeback/deciding-set/tiebreak resilience against high-quality opposition.

A qualifying ceiling event triggers mandatory escalation. It cannot be dismissed merely because the favorite leads aggregate Elo, market, Monte Carlo, or model vote count.

### 5. PRESENT-STRENGTH / RATING-LAG TEST

Compare long-run ratings against last 30/60/90-day and same-surface performance. Explicitly classify each player:

CURRENTLY REPRESENTATIVE / POSSIBLY STALE / CLEARLY LAGGING.

If the underdog's rating is lagging rapid improvement, historical Elo/ranking receives reduced decision authority.

### 6. ELITE-FAVORITE EXEMPTION PROHIBITED

No 80%, 85%, 90%+ favorite is exempt from the full underdog audit. High headline probability, Strong Agreement, Elite recommendation, or high Monte Carlo cannot suppress a demonstrated dangerous-underdog signal.

### 7. H2H QUALITY CONTROL

H2H must be weighted for recency, surface, tour level, age/development stage, current player level, and sample size. Old/thin/non-comparable H2H cannot veto stronger present-strength evidence.

## 8. CORRELATED AGREEMENT COMPRESSION

Surface Elo, General, Specialist, Monte Carlo, headline probability, and market may share inputs. Count effective independent evidence families, not raw agreeing metrics. Strong Agreement is not sufficient for Green if effective independent evidence is low.

## 9. GREEN / DOUBLE-GREEN EXIT STANDARD

Before Green, the favorite must:

- defeat the underdog's 5 strongest pre-match arguments;
- survive the conservative probability stress range;
- survive removal of the strongest favorite component;
- have no unresolved Dangerous-Underdog ceiling trigger;
- have sufficient CRITICAL evidence coverage;
- survive updated historical calibration.

Double Green requires all Green conditions plus exceptionally strong independent evidence and no material unresolved contradiction.

## 10. POST-RESULT LEARNING LEDGER

Every completed match must be appended to the verified calibration. For every Green/Double-Green loss, classify the pre-match failure mechanism using only evidence available before first serve. For every successful downgrade/removal and every false downgrade/flip, record why. Recurrent structures become future downgrade or anti-overreaction rules.

## VERIFICATION-SPECIFIC ADDITIONS

### A. MANDATORY UNDERDOG PRESENT-STRENGTH PACKET

For every match, independently verify the underdog's last 3/5/10 matches, same-surface results, opponent quality, best recent wins, competitive losses, ranking/Elo trajectory, serve/return trend, workload, and availability. Do this even for 85%+ favorites.

### B. CEILING-EVENT SEARCH

Search specifically for the underdog's strongest recent win(s) and strongest recent performance(s), not merely average form. Identify whether these show a level that long-run ratings have not yet absorbed.

### C. FAVORITE VULNERABILITY MATCH

Take the favorite's recent loss archetypes and test whether this exact underdog has demonstrated the tools to reproduce them. Generic "favorite is stronger" language cannot close this test.

### D. STALENESS / RECENCY RECONCILIATION

When career/52-week numbers disagree with 30/60/90-day or current-surface evidence, show both and explain which deserves more weight. Do not silently average away a rapid rise or decline.

### E. CRITICAL DATA COMPLETENESS

Present strength, current surface form, serve/return, opponent quality, physical availability, and recent ceiling events are CRITICAL. A Green cannot be issued if a critical family is genuinely unresolved unless the remaining independent evidence is overwhelming and the uncertainty is explicitly stress-tested.

# 120-MATCH EMPIRICAL VERIFICATION WEIGHTING UPDATE

This section is additive and sits above the original Verification Audit. It changes evidence priority and Green/Yellow/Red gating without deleting any existing verification requirement.

## A. Empirical evidence hierarchy

**Primary empirical separators:** Elo separation, Matchup Closeness, Monte Carlo support, meaningful recent-form differential after opponent adjustment, Surface Elo support with valid sample/reliability, and current market confirmation when available.

**Secondary/contextual:** fatigue/workload, ranking trajectory, pressure/closing, common opponents, style, straight-set call, and recommendation labels.

**Weak standalone evidence:** H2H, Data Quality, Upset Risk, Strong Agreement, raw model-vote count, and one recent-form contradiction.

## B. Green gate changes

Close / coin-flip structure is a major Green blocker because the current sample was 14/27 (51.9%). Elo gap <50 is downgrade pressure because the current sample was 32/52 (61.5%), compared with 43/53 (81.1%) at Elo gap >=50.

MC <55% is downgrade pressure (14/22, 63.6%). MC >=55% is supportive (61/83, 73.5%). MC >=70% is strong but currently small-sample (11/12).

A Recent Form advantage >10 was 30/38 (78.9%), but Recent Form must remain opponent-adjusted and cannot independently force a flip.

Surface Elo support >=70% was 23/28 (82.1%). Specialist support <65% was 12/22 (54.5%) and should trigger a deeper contradiction review.

## C. Independence and correlation enforcement

General Model and Specialist Model are treated as potentially highly correlated with Surface Elo/headline/MC. Agreement across these outputs increases confidence only after independent evidence families confirm it. Do not award multiple full-strength votes for overlapping outputs.

## D. Empirical combination tests

**Elo gap  $\geq 50$  + MC  $\geq 55$ :** 40/49 = 81.6%. **Elo gap  $\geq 50$  + recent-form advantage  $> 10$ :** 17/20 = 85.0%. **Market  $\geq 60$  + Elo gap  $\geq 50$ :** 10/11 = 90.9% (small sample). **Close + MC  $< 55$ :** 7/13 = 53.8%. **Close + Elo gap  $< 50$ :** 11/21 = 52.4%.

## E. Final classification changes

**DOUBLE GREEN:** requires strong independent-family agreement, no Close/Coin-Flip structure, successful component-removal stress test, no dangerous-underdog ceiling trigger, and strong empirical calibration support. Small-sample combinations may support but never create Double Green alone.

**GREEN:** favorite remains favored after opponent adjustment and removal of its strongest family; no multiple independent underdog pathways; empirical structure is not in a demonstrated weak zone unless overwhelming independent evidence survives stress testing.

**YELLOW:** use when Elo separation is thin, MC is weak, closeness is dangerous, current form conflicts with long-run strength, or one meaningful opponent-specific pathway remains.

**RED/PASS:** use when multiple independent reliable families contradict the favorite, or weak empirical structure combines with a demonstrated underdog pathway. RED/PASS does not automatically mean FLIP.

# ORIGINAL AUDIT — PRESERVED IN FULL

Tennis Matrix AI - Reusable MAX Metrics Plus Audit

Page 1

TENNIS MATRIX AI

REUSABLE FULL VERIFICATION AUDIT

MAX METRICS PLUS - PUBLIC / RECONSTRUCTABLE EDITION

MASTER TEMPLATE - NO MATCHES INCLUDED

Expanded with advanced charting, score-state, opponent-adjusted residual, volatility, floor/ceiling, archetype, current-tournament, schedule, environmental, and opponent-specific cross-feature layers. Use only public or transparently reconstructed inputs.

EXECUTION RULES - READ BEFORE EVERY AUDIT

Execute the entire audit exactly as written for every matchup. Search and reconstruct every applicable public metric. Do not skip a metric because it is time-consuming. DIRECT, RECONSTRUCTED, PARTIAL, UNAVAILABLE and EXCLUDED must be distinguished. Do not invent missing observations. Do not produce the final color list until every matchup has passed the completion ledger.

The audit's job is not to prove the Tennis Matrix prediction correct. Its job is to attempt to falsify it. Start from a neutral position, construct the strongest evidence-based case for both competitors, and award Green only when the Matrix winner survives the attempt to disprove it.

ENFORCEMENT RULES 1-60

1. No Metric Counting Rule - 20 metrics favoring Player 1 does not automatically beat 8 favoring Player 2. Correlated statistics cannot be treated as separate independent votes.
2. Evidence-Family Rule - Collapse evidence into independent families before deciding: Strength/Elo, Serve, Return, Form, Surface, Opponent-Specific Matchup, Pressure/Closing, Workload, Tactical/Charting, Market, Availability, and Current-Tournament evidence.
3. Reliability Beats Quantity Rule - One HIGH-reliability contradiction can outweigh several LOW-reliability supporting metrics. Every decisive metric gets HIGH/MEDIUM/LOW reliability and FULL/REDUCED/CONTEXT/ZERO weight.
4. Sample Floor Rule - Small samples cannot receive full weight. Surface splits, H2H, tiebreaks, deciding sets, archetype results, charting, and environmental splits need explicit sample checks.
5. Recency x Sample Balance Rule - Do not automatically prefer recent or career data. Recent evidence is relevant but noisy; large historical samples are stable but can be stale. Reconcile both.
6. Opponent-Adjustment Requirement - Raw form cannot be decisive until opponent strength is considered.
7. Surface Requirement - Overall numbers cannot override strong surface-specific evidence without explanation.
8. Tour-Level Translation Rule - ITF -> Challenger -> ATP/WTa performance must be adjusted for competition level.
9. No H2H Worship Rule - Reduce H2H weight when old, on another surface, based on few meetings, or when player level/style has materially changed.
10. No Tiebreak/Clutch Worship Rule - Strong tiebreak, break-point, or deciding-set records must be tested for regression before being considered durable skill.
11. Underlying Performance Rule - Results and scorelines must be checked against points, games, hold/break, and opponent strength.
12. Winner/Loss Quality Rule - Distinguish strong wins, weak wins, strong losses, and bad losses.
13. Scoreline Deception Rule - Never infer dominance from 2-0/3-0 alone. Check games, points, breaks, and pressure situations whenever available.
14. Current-Tournament Override Check - Current tournament serve, return, movement, workload, and conditions receive a dedicated check against historical profile.
15. Skill-Divergence Rule - Recent Form cannot remain one number. Determine whether improvement/decline comes from serve, return, both, or pressure-point variance.
16. Matchup Interaction Rule - Explicitly ask whether one player possesses the exact weapon that attacks the opponent's documented weakness.
17. Bidirectional Matchup Rule - Every interaction must be tested in both directions.

18. Failure-Mode Replication Rule - Identify how the favorite has actually been losing and determine whether today's opponent can reproduce those conditions.
19. Underdog Pathway Minimum Rule - Major trap concern requires one exceptionally strong pathway or multiple independent credible pathways.
20. Favorite Floor Rule - Green requires analysis of the favorite's bad-day performance, not only average or ceiling performance.
21. Underdog Ceiling Rule - Analyze the underdog's credible upper range, not only average performance.
22. Distribution Before Point Estimate Rule - Report uncertainty and performance-distribution overlap rather than relying only on headline probabilities.
23. Uncertainty Penalty Rule - Missing data, lower-tour identity uncertainty, surface transitions, small samples, and high volatility must reduce confidence.
24. Missing-Data Asymmetry Rule - Check whether missing information disproportionately affects one player. Sparse data cannot silently benefit the better-known favorite.
25. Charting Coverage Rule - Tactical charting cannot receive full weight without reporting charted matches, surfaces, dates, and opponent quality.

Tennis Matrix AI - Reusable MAX Metrics Plus Audit

Page 2

26. Market Independence Rule - Market information is validation, not automatically truth. Multiple books are not automatically multiple independent signals.
27. Market Movement Context Rule - Line movement requires magnitude, timing, source/liquidity quality, and corroborating tennis evidence before receiving major weight.
28. No Double-Counting Matrix Outputs Rule - General, Specialist, Surface Elo, Recent Form, and other outputs must not be assumed independent when their inputs may overlap.
29. Unknown Correlation Penalty Rule - When exact model overlap is unknown, classify independence as UNKNOWN and reduce confidence in apparent unanimity.
30. Reconstruction Transparency Rule - Every reconstructed number must show the formula or method, inputs, and source observations.
31. No Reconstruction From Reconstruction Rule - Avoid repeatedly stacking uncertain derived estimates into additional metrics that masquerade as independent evidence.
32. Source Hierarchy Rule - Prefer official tour/tournament statistics, then established statistical databases, reputable reporting, and only then lower-authority contextual sources.
33. Cross-Source Verification Rule - Critical facts such as withdrawals, injuries, match status, and surprising statistics should be checked against a second source whenever practical.
34. Identity Verification Rule - Especially for Challenger/ITF players, verify identity before joining statistics.
35. Data Timestamp Rule - Record when data was retrieved and whether it includes the player's most recent match.
36. No Future Leakage Rule - In retrospective audits, only information available before the match began may be used.
37. Pre-Match Lock Rule - For live predictions, establish a cutoff timestamp. Information after first serve cannot retroactively improve the prediction.
38. Retirement/Walkover Rule - Separate normal completed matches from retirements, walkovers, and abnormal matches when calculating form/performance.
39. Injury Evidence Rule - Do not infer injury from poor performance alone. Injury requires credible public evidence.
40. Outlier Rule - One extreme match cannot dominate the conclusion. Test results with and without major outliers.
41. Sensitivity Rule - Rerun the conclusion after reducing/removing each major evidence family. If the winner flips easily, the pick is fragile.
42. Winner-Switch Rule - For reconstructed models, estimate how much Elo, hold%, break%, form, or market probability must change before the projected winner switches.
43. Leave-One-Family-Out Rule - Remove Serve, Return, Elo, Form, Market, etc. one family at a time and test whether the conclusion survives.
44. Adversarial Underdog Rule - Intentionally construct the strongest evidence-based case for the underdog, then try to disprove it.
45. Adversarial Favorite Rule - Do the reverse for the favorite. The audit must be capable of defending whichever side survives scrutiny.
46. Contradiction Severity Rule - Classify contradictions as MINOR / MODERATE / MAJOR / CRITICAL.
47. Unresolved Contradiction Rule - A CRITICAL unresolved contradiction automatically blocks Double Green.
48. Double Green Qualification Rule - Double Green requires strong independent-family agreement, adequate samples, low distribution overlap, no major availability concern, no strong opponent-specific counter, and survival of stress testing.
49. Green Qualification Rule - Green requires the favorite to remain favored after opponent adjustment and reasonable stress tests, with no multiple independent high-quality underdog pathways.

50. Automatic Yellow Rule - Yellow if evidence is genuinely split, sample quality is poor, distribution overlap is substantial, current form conflicts with long-term strength, or one

major opponent-specific vulnerability remains unresolved.

51. Automatic Red Rule - Red when multiple independent reliable evidence families contradict the Matrix favorite, or a documented favorite weakness aligns strongly with an

opponent strength and survives reliability checks.

52. PASS Rule - PASS is valid even if the Matrix says 80%. Confidence in the Matrix number and confidence in independent verification are separate.

53. No Forced Winner Rule - If evidence cannot distinguish the players reliably, say so. Verification is filtering, not manufacturing action.

54. Final Probability Reconciliation Rule - Report Matrix probability, independent reconstructed probability/range, and market no-vig probability separately.

55. Probability-Gap Alarm - Large Matrix vs reconstructed vs market disagreement automatically triggers deeper investigation before Green.

56. Confidence Ceiling Rule - Final confidence cannot exceed the reliability of the evidence supporting it.

57. Audit Completion Gate - No final color until every mandatory section is COMPLETE, PARTIAL with explanation, UNAVAILABLE after attempted sourcing, or legitimately

EXCLUDED.

58. No Shortcut Declaration - Final output includes a completion percentage and explicitly lists unavailable sections.

59. Evidence Ledger Rule - Finish each match with: Evidence Family | P1 | P2 | Reliability | Sample | Winner | Contradiction? | Weight.

60. Decision Trace Rule - Final classification must state exactly which 3-5 independent evidence families drove the decision and what evidence could realistically overturn it.

Tennis Matrix AI - Reusable MAX Metrics Plus Audit

Page 3

Web-Reconstructable / Public-Source Edition

Purpose: Preserve the original verification structure and match-by-match Matrix evidence while removing the specific private, proprietary, physiological, tracking, and unavailable-source items identified for deletion. Replace those items only with metrics an AI can reconstruct from the supplied Matrix summary/export or retrieve/reconstruct from current public web sources.

Allowed evidence families in this edition

■ Matrix-export fields actually supplied: headline probability, recommendation, upset risk, data quality, model agreement, closeness, component votes, reliability/sample/status, Surface Elo output, Serve/Return output, Recent Form output, H2H output, Market output, General/Specialist output, fatigue index if exported, Monte Carlo if exported, set-score call if exported.

■ Public ranking/form: current ranking, ranking movement, last 5/10, streak, same-surface record, tournament results, quality wins, bad losses, opponent quality.

■ Public serve/return statistics: hold/break rates, service/return points won, first/second-serve performance, aces, double faults, break points saved/converted, where a reliable source publishes them.

■ Public match-log reconstruction: Set 1 outcomes, deciding sets, tiebreaks, comeback/closing patterns, scoreline progression, common opponents, surface transitions, tournament workload, qualifying load, rest intervals.

■ Public availability/context: injuries, withdrawals, retirements, medical timeouts, return-from-injury reports, coaching/equipment changes, venue history, draw status, lucky-loser/walkover status.

■ Public market validation: multi-book moneyline consensus, no-vig probability, opening/current/closing movement when obtainable, and prediction-market information when available.

■ Reconstructable matchup proxies: expected hold/break from public serve/return data, second-serve attack mismatch, common-opponent differential, surface-adjusted recent form, workload differential, and documented tactical/style interaction.

■ Public historical analogs: comparable public matches selected from reconstructable dimensions. Do not claim they are the private 100-match workbook, and do not invent proprietary similarity scores.

Explicitly removed

The edition does not require or claim access to: an independent private Green list; hidden General/Specialist features, weights, calibration or exact overlap; proprietary internal indices/formulas; the unsupplied private 100-match workbook or its exact trap twins; private physiological measurements; or instrumented player/ball tracking data.

## Operating rule

Missing public data is marked UNAVAILABLE or PARTIAL. It is never guessed. Original Matrix outputs may be reported as outputs, but hidden model internals are not reverse-engineered or invented. Any original final color retained below is an original audit classification unless a fresh public-source rerun is explicitly performed.

### MANDATORY MASTER METRICS SWEEP

Non-negotiable rule: Before issuing DOUBLE GREEN, GREEN, YELLOW, RED, or PASS, search for or reconstruct every applicable metric below for both players. A metric may be used only when it is (A) present in the supplied Tennis Matrix export, (B) obtainable from a current/reliable public source, or (C) transparently reconstructed from public inputs. Never invent missing data.

For every applicable metric use this comparison chain:

Player 1 value -> Player 2 value -> direct differential -> surface-adjusted differential -> recent differential -> opponent-adjusted differential -> reliability/sample -> evidence direction.

Every field must be labeled:

- DIRECT - published/supplied value.
- RECONSTRUCTED - calculated transparently from public inputs.
- PARTIAL - some required inputs are missing.
- UNAVAILABLE - cannot be obtained reliably.
- EXCLUDED - supplied but unusable by status/reliability.

Tennis Matrix AI - Reusable MAX Metrics Plus Audit

Page 4

### A. Ranking / Rating / Opposition Strength

- Current ATP/WTB ranking.
  - Ranking points.
  - Ranking movement: 1/4/8/12/26/52 weeks where reconstructable.
  - Career-high ranking.
  - Ranking trajectory and velocity.
  - Ranking differential.
  - Elo / overall Elo when publicly available.
  - Surface Elo when publicly available.
  - Elo differential.
  - Elo-implied win probability.
  - Recent Elo trend.
  - Peak Elo.
  - Average opponent ranking.
  - Average opponent Elo where available.
  - Strength of schedule.
  - Opponent-quality percentile.
  - Tour-level-adjusted strength.
  - Performance vs Top 10/20/50/100/200/500.
  - Upset win rate.
  - Loss rate to lower-ranked opponents.
  - Ranking-performance disconnect.
  - Ranking-lag / rapid-riser proxy.
  - Age/development trajectory.
- ### B. Surface / Conditions
- Career hard/clay/grass record and win%.



- Last-52-week surface record.
- Last 20/10/5 matches on surface.
- Surface Elo and differential.
- Surface hold%.
- Surface break%.
- Surface service-points-won%.
- Surface return-points-won%.
- Surface first-serve points won%.
- Surface second-serve points won%.
- Surface first-serve return points won%.
- Surface second-serve return points won%.
- Surface tiebreak%.
- Surface deciding-set%.
- Surface opponent quality.
- Surface quality-win rate.
- Surface bad-loss rate.

Tennis Matrix AI - Reusable MAX Metrics Plus Audit

Page 5

- Indoor vs outdoor.
- Fast/medium/slow court split when court-speed classification is public.
- Altitude record/context.
- Tournament-specific record.
- Venue history.
- Surface-transition record.
- First match after surface switch.
- Surface-adjusted recent form.
- Weather/temperature/humidity/wind when relevant and publicly obtainable.
- Ball type/court speed only when reliably documented.

#### C. Serve

- Ace%.
- Aces per match/set/service game.
- Double-fault%.
- Double faults per match/set/service game.
- Ace-to-double-fault ratio.
- First-serve-in%.
- First-serve points won%.
- Second-serve points won%.
- Service points won%.
- Hold/service-games-won%.
- Break-points-saved%.
- Break points faced per service game.
- Unreturned serve%.
- Serve winner%.
- Service in-play points won%.
- First-serve effectiveness.

- Second-serve effectiveness.
- Serve rating where source formula is known.
- Clean-hold rate when reconstructable.
- Love-hold rate when reconstructable.
- Hold after facing break point.
- Hold after 0-30 / 15-30 when point logs support it.
- Serve-out-set conversion.
- Serve-out-match conversion.
- First-serve performance under pressure.
- Second-serve performance under pressure.
- Serve deterioration by set.
- First-serve percentage by set.
- Ace rate by set.
- Double-fault rate by set.
- Hold% by set.

Tennis Matrix AI - Reusable MAX Metrics Plus Audit

Page 6

- First/second serve speed where published.
- Serve-speed trend where published.
- First-to-second serve speed ratio where published.
- Maximum-to-average serve speed ratio where published.
- Serve +1 / <=3-shot win% when charting exists.
- Serve Impact when charting exists.

#### D. Return

- Return points won%.
- Return games won / break%.
- First-serve return points won%.
- Second-serve return points won%.
- Break-point conversion%.
- Break points created per return game.
- Return-in-play%.
- Return-in-play win%.
- Return winner%.
- Return unforced-error% where charted.
- Return depth / Return Depth Index where charted.
- Slice/chip return% where charted.
- Ace-against%.
- Opponent double-fault rate.
- Clean-break rate when reconstructable.
- Love-break rate when reconstructable.
- Return pressure frequency.
- Break opportunity frequency.
- Second-serve attack effectiveness.
- Return performance by surface.
- Return performance vs comparable servers.

■ Return performance by opponent handedness where sample permits.

■ Break-back rate.

■ Immediate rebreak rate.

■ Return +1 effectiveness where charted.

#### E. Opponent-Specific Serve-vs-Return Interaction

■ P1 hold% vs P2 break%.

■ P2 hold% vs P1 break%.

■ Expected hold for each player.

■ Expected break for each player.

■ Expected hold differential.

■ Expected break differential.

■ P1 first-serve strength vs P2 first-serve return.

■ P2 first-serve strength vs P1 first-serve return.

■ P1 second serve vs P2 second-serve return.

#### Tennis Matrix AI - Reusable MAX Metrics Plus Audit

Page 7

■ P2 second serve vs P1 second-serve return.

■ Second-serve mismatch differential.

■ Ace production vs ace suppression.

■ Double-fault vulnerability vs return pressure.

■ Break-point saving vs break-point conversion.

■ Serve dominance vs return strength.

■ Projected service-game advantage.

■ Opponent-specific serve/return residual.

■ Expected number of breaks.

■ Projected tiebreak likelihood.

All interaction metrics must expose the inputs/formula and be labeled RECONSTRUCTED.

#### F. Recent Form / Momentum

■ Last 3/5/10/20 record.

■ Last-52-week record.

■ Current streak.

■ Recent sets won/lost.

■ Recent games won/lost.

■ Recent point differential where available.

■ Form score.

■ Recency-weighted form.

■ Opponent-adjusted form.

■ Surface-adjusted form.

■ Opponent-and-surface-adjusted form.

■ Quality of last 3/5/10 opponents.

■ Quality wins.

■ Bad losses.

■ Average ranking/Elo beaten.

■ Average ranking/Elo lost to.

■ Expected vs actual wins.

- Straight-set win%.
- Straight-set loss%.
- Games won%.
- Sets won%.
- Total-points-won%.
- Hold trend.
- Break trend.
- Serve trend.
- Return trend.
- Performance acceleration/deceleration.
- Improvement despite losses.
- Decline despite wins.

#### G. Favorite Loss Autopsy

Tennis Matrix AI - Reusable MAX Metrics Plus Audit

Page 8

For recent relevant losses reconstruct:

- Opponent.
- Opponent ranking/rating/quality.
- Surface.
- Tournament level.
- Score.
- Set 1 result.
- Won Set 1 then lost?
- Break lead surrendered?
- Deciding-set loss?
- Tiebreak loss?
- Serve decline.
- Return decline.
- Second-serve weakness.
- Double-fault spike.
- Break-point failures.
- Break-consolidation failures.
- Serve-out failures.
- Competitive vs blowout.
- Match duration/workload.
- Opponent archetype.
- Repeatable tactical failure pattern.
- Loss frequency to lower-ranked players.
- Repeated loss archetypes.

#### H. Predicted-Loser Win Autopsy

For strongest/relevant recent wins reconstruct:

- Opponent quality.
- Opponent ranking/rating.
- Surface.
- Tournament level.

- Score.
- Whether opponent was higher ranked/favored.
- Straight-set vs deciding-set.
- Comeback after losing Set 1.
- Deciding-set win.
- Tiebreak win.
- Serve dominance.
- Return dominance.
- Break differential.
- Point dominance.
- Second-serve attack.
- Lead protection.

Tennis Matrix AI - Reusable MAX Metrics Plus Audit

Page 9

- Win over comparable favorite.
- Repeatability of winning pathway.
- Quality-win frequency.

#### I. Point / Game / Set Dominance

- Total points won%.
- Service points won%.
- Return points won%.
- Games won%.
- Service games won%.
- Return games won%.
- Sets won%.
- Matches won%.
- Points Dominance Ratio.
- In-play Points Dominance Ratio.
- Second-serve in-play dominance.
- Games Dominance Ratio.
- Break Points Ratio.
- Point-to-game conversion.
- Point-to-set conversion.
- Point-to-match conversion.
- Service-points-to-service-games overperformance.
- Return-points-to-return-games overperformance.
- Points-to-tiebreak overperformance.
- Break-points overperformance.
- Break-points-saved overperformance.
- Break-points-converted overperformance.
- Dominance vs opponent quality.

#### J. Set 1 / Start Speed

- Set-1 win%.
- First-break frequency.
- First break conceded%.

- Opening service-game hold%.
- Opening return-game break%.
- First-four-games differential.
- First-six-games differential.
- Early-break frequency.
- Early-break recovery%.
- Set-1 tiebreak%.
- Set-1 point/game differential.
- Games/time until first break where reconstructable.
- Performance after holding first service game.
- Performance after being broken first.

Tennis Matrix AI - Reusable MAX Metrics Plus Audit

Page 10

- Slow-start frequency.
- Fast-start frequency.

K. Comeback / Closing / Lead Durability

- Win% after winning Set 1.
- Win% after losing Set 1.
- Set-2 response after losing Set 1.
- Deciding-set win%.
- Comeback win%.
- Loss after winning Set 1%.
- Break-back%.
- Break-consolidation%.
- Serving-for-set conversion%.
- Serving-for-match conversion%.
- Lead durability.
- Break-lead surrender rate.
- Double-break surrender.
- Opponent-serving-for-match break rate.
- Performance from break down.
- Performance from break ahead.

L. Tiebreak / Pressure / Clutch Sustainability

- Career tiebreak%.
- Recent tiebreak%.
- Surface tiebreak%.
- Tiebreak frequency.
- First-set tiebreak%.
- Deciding-set tiebreak%.
- Tiebreak point differential where chartable.
- Expected-vs-actual tiebreak performance.
- Break-points saved%.
- Break-points converted%.
- Break opportunities per return game.
- Break points faced per service game.

- Deuce-game performance when reconstructable.
- Break consolidation.
- Break-back.
- Serve-out rates.
- Pressure hold%.
- Pressure return%.
- Deciding-set performance.
- Clutch over/underperformance relative to underlying point strength.

#### M. Match Length / Endurance Proxies

Tennis Matrix AI - Reusable MAX Metrics Plus Audit

Page 11

- Average match duration.
- Recent minutes on court.
- Recent games played.
- Recent sets played.
- Recent three-setters/five-setters.
- Consecutive long matches.
- Deciding-set frequency.
- Record in long matches.
- Performance by set number.
- Third-set hold/break%.
- Fifth-set performance when applicable.
- Late-set performance.
- Serve deterioration later in matches.
- Return improvement/decline later in matches.

#### N. Workload / Recovery Proxies

- Matches in last 3/7/14/30 days.
- Sets in last 7/14/30.
- Games in last 7/14/30.
- Public minutes on court.
- Consecutive-day play.
- Qualifying workload.
- Matches this tournament.
- Recent three-/five-setters.
- Late finishes.
- Hours since previous match when reconstructable.
- Tournament-to-tournament travel.
- Time-zone change.
- Surface change.
- Consecutive tournaments.
- Walkover/rest advantage.
- Bye/rest advantage.

These are observable workload proxies only; never convert them into claims about private biological fatigue.

#### O. H2H / Rematch

- Overall H2H.

- Surface H2H.
- Recent H2H.
- H2H set record.
- H2H game differential.
- Straight-set frequency.
- Deciding-set meetings.
- Tiebreak meetings.
- First-break patterns where reconstructable.

Tennis Matrix AI - Reusable MAX Metrics Plus Audit

Page 12

- Prior-meeting serve/return performance.
- Score progression.
- Rematch turnaround.
- Time since previous meeting.
- Same-tournament/venue H2H.
- Junior/ITF H2H when reliable.
- Documented tactical adjustments.

P. Common Opponents / Network Evidence

- Common-opponent W/L.
- Same-surface common opponents.
- Recent common opponents.
- Score differential.
- Set differential.
- Game differential.
- Point differential where available.
- Hold differential.
- Break differential.
- Performance against similar ranking.
- Performance against similar Elo.
- Performance against similar archetypes.
- Recency-weighted common-opponent score.
- Opponent-strength-weighted common-opponent score.
- Second-degree common opponents.
- Transparent transitive-performance estimate.
- Transparent network-strength estimate.

Do not present reconstructed network scores as proprietary Tennis Matrix metrics.

Q. Rally / Shot / Charting Metrics - only when public charting exists

- Rally length.
- 0-4 shot performance.
- 5-8 shot performance.
- 9+ shot performance.
- Short-rally win%.
- Medium-rally win%.
- Long-rally win%.
- First-strike efficiency.



- Serve +1.
- Return +1.
- Forehand winner/error rates.
- Backhand winner/error rates.
- Winners.
- Unforced errors.
- Forced errors.

Tennis Matrix AI - Reusable MAX Metrics Plus Audit

Page 13

- Winner-to-unforced-error ratio.
- Net approach frequency.
- Net points won%.
- Net effectiveness.
- Passing-shot success.
- Drop-shot frequency/success.
- Slice frequency.
- Return depth.
- Serve placement.
- Return position.
- Direction patterns.

Coverage must be checked; charted samples may be small and non-representative.

R. Style / Tactical Matchup

- Handedness.
- One- vs two-handed backhand.
- Big server vs elite returner.
- Aggressive baseliner vs counterpuncher.
- First-strike vs rally tolerance.
- Net attacker vs passing ability.
- Second-serve vulnerability vs aggressive returner.
- Weak-wing exposure where documented.
- Pace generation vs pace absorption.
- Surface/style compatibility.
- Return-position compatibility.
- Short-rally vs long-rally advantage.
- Rematch tactical adaptability.
- Documented preferred-pattern collision.

Do not invent subjective style labels when reliable evidence is absent.

S. Market / Price Validation

- Current moneyline.
- Implied probability.
- No-vig probability.
- Multi-book consensus.
- Opening probability.
- Current probability.
- Closing probability when available.

- Line movement.
- Magnitude of movement.
- Favorite drift/steam.
- Underdog shortening.
- Book disagreement/dispersion.

## MANDATORY MARKET CONSENSUS ENFORCEMENT - FULL

### VERIFICATION AUDIT

This page is part of the audit, not an optional appendix. It strengthens S. Market / Price Validation, the Probability-Gap Alarm, Market-Veto Investigation, stress testing, and Final Green eligibility.

1. Sportsbook source minimum. For every current match, attempt at least 3 independent sportsbooks when available. Prefer one sharp/reference or high-liquidity book plus two mainstream regulated books. An odds aggregator may help locate prices, but it does not count as multiple independent books unless the underlying books are individually identified.
2. Same-market / same-time rule. Capture both players from the same match-winner market and the same book at approximately the same timestamp. Record sportsbook, timestamp, favorite price, underdog price, and source URL. Suspended, stale, one-sided, or mismatched-round prices are not valid consensus inputs.
3. De-vig each sportsbook separately. Convert both sides to raw implied probability, then compute the book-level no-vig favorite probability as  $q_{Fav} / (q_{Fav} + q_{Dog})$ . Build the live sportsbook consensus from the median no-vig probability across valid books; also record low/high range and dispersion.
4. Prediction markets are separate. Kalshi and other prediction markets must be recorded as a separate evidence family. Never treat Kalshi American-style display odds as ordinary sportsbook moneylines, never blend them into sportsbook consensus, and never use them as a substitute for a two-sided sportsbook quote.
5. Kalshi coherence check. When Kalshi is used, prefer the underlying contract price/probability and liquidity/spread information. If its displayed American-style numbers produce an extreme synthetic overround, cannot be reconciled with complementary contract probabilities, or materially conflict with the underlying contract price, label the display NON-COMPARABLE / DISPLAY-STRUCTURE WARNING and give it no sportsbook-consensus weight.
6. Market integrity thresholds. A sportsbook-consensus vs Matrix-market gap of 5+ percentage points requires investigation; 8+ pp is a MATERIAL MARKET CONFLICT; 12+ pp is a CRITICAL MARKET CONFLICT and blocks Green/Keep until explained. Apply the same 5/8/12 pp ladder to Kalshi vs sportsbook consensus, but treat the result as a prediction-market discrepancy rather than sportsbook evidence.
7. Book-dispersion warning. If valid sportsbook no-vig probabilities span 8+ pp, mark MARKET DISPERSION HIGH and do not claim a precise consensus. Investigate stale lines, limits/liquidity, player identity, tournament/round, withdrawals, and news before assigning decision weight.
8. Source-count downgrade. Three or more valid books = COMPLETE. Two valid books = PARTIAL and reduced market weight. One valid book = SINGLE-SOURCE ONLY and cannot independently unlock Green. Zero reliable books = CURRENT SPORTSBOOK MARKET UNAVAILABLE.
9. No market double-counting. Matrix Market Consensus, live sportsbook consensus, and prediction-market information must remain three distinct fields. Agreement may strengthen confidence; disagreement must be explained. Do not count multiple books as multiple independent tennis signals; they form one Market evidence family.
10. Market cannot be silently ignored. A material/critical conflict must appear in the contradiction ledger, trap score, stress tests, removal case, and final decision rationale. If unresolved, downgrade/remove/pass rather than certifying Green.

Mandatory per-match market ledger

Field

Required entry

Sportsbooks

Book 1 / Book 2 / Book 3+; prices; timestamp; URL

No-vig consensus

Median; low-high range; dispersion; valid source count

Matrix market

Separate Matrix Market Consensus probability

Prediction market

Kalshi/other contract probability, liquidity/spread; separate

Conflict flags

Matrix-vs-sportsbook pp gap; prediction-vs-sportsbook pp gap

Integrity status

COMPLETE / PARTIAL / SINGLE-SOURCE / UNAVAILABLE

Decision effect

CONFIRMS / NEUTRAL / WEAKENS / MATERIAL CONFLICT / CRITICAL VETO

VERIFICATION GREEN VETO: A MATERIAL conflict requires explicit resolution before Green. A CRITICAL conflict automatically blocks Green/Double-Green until the cause is identified and the favorite still survives the reconstructed evidence after the market discrepancy is accounted for.

Tennis Matrix AI - Reusable MAX Metrics Plus Audit

Page 14

- Matrix-vs-market gap.
- Elo-vs-market gap.
- Market-vs-form gap.
- Price-bucket historical performance when reconstructable.
- Closing-line accuracy when historical prices exist.
- Prediction-market probability when available.

T. Tournament / Venue / Match Context

- Tournament level.
- Round.
- Qualifying vs main draw.
- Qualifier status.
- Lucky-loser status.
- Wildcard status.
- Protected-ranking status.
- Seed status.
- Defending-points context.
- Previous-year result.
- Tournament history.
- Venue history.
- Home-country performance.
- Altitude.
- Indoor/outdoor.
- Court speed when documented.
- Ball type when documented.
- Weather.
- Schedule time/day-night.
- Current draw validity.
- Opponent replacement.

- Walkover status/history.

- Rest advantage.

#### U. Public Availability / Physical Evidence

Use only public/observable information:

- Reported injury.

- Reported body area.

- Withdrawal.

- Retirement.

- Medical timeout.

- Illness.

- Layoff length.

- Matches since return.

- Record since return.

- Serve-speed decline where published.

#### Tennis Matrix AI - Reusable MAX Metrics Plus Audit

Page 15

- First-serve/hold decline after return.

- Publicly documented movement concern.

- Repeated retirements/withdrawals.

- Current participation status.

Never claim private sleep, hydration, soreness, pain, treatment, medication, physiological recovery, muscle fatigue, cardiovascular fatigue, or private training load.

#### V. Stability / Regression / Variance

- Tiebreak overperformance.

- Deciding-set overperformance.

- Break-point overperformance.

- Match-win% vs point-win% overperformance.

- Set-win% vs point-win% overperformance.

- Game-win% vs point-win% overperformance.

- Clutch sustainability.

- Recent win% vs underlying point%.

- Recent record vs Elo expectation.

- Ace/double-fault regression.

- Break-point conversion regression.

- Break-point save regression.

- Opponent conversion variance.

- Strength-of-schedule adjustment.

- Luck/variance indicator.

#### W. Transparent Matchup Probability Reconstruction

When enough public inputs exist, independently reconstruct:

- Elo probability.

- Surface-Elo probability.

- Ranking-derived baseline.

- Hold/break Markov estimate.

- Serve/return probability estimate.

- Recent-form estimate.
- Surface-form estimate.
- Common-opponent estimate.
- Market no-vig probability.
- Transparent ensemble estimate.
- Monte Carlo from public hold/break assumptions.
- Straight-set probability.
- Deciding-set probability.
- Expected sets.
- Expected games.
- Tiebreak probability.
- Upset probability.
- Uncertainty interval.

Tennis Matrix AI - Reusable MAX Metrics Plus Audit

Page 16

Every result here must be labeled RECONSTRUCTED OUTSIDE ESTIMATE, never a Tennis Matrix proprietary output.

#### X. Reliability / Sample / Source Quality

For every metric record:

- Matches in sample.
- Surface matches in sample.
- Service/return points or games in sample when known.
- Recency.
- Opponent quality.
- Source coverage.
- Charted-match coverage.
- Missing-stat rate.
- Confidence interval where calculable.
- Effective sample size where calculable.
- Reliability grade.
- Cross-source agreement.
- Player-identity confidence, especially lower tours.
- Stale-data flag.
- Retirement-distorted sample flag.
- Small-sample warning.

#### MASTER METRICS ENFORCEMENT

1. Do not stop after finding enough evidence to support the Matrix favorite.
2. Player 1 and Player 2 receive the same metric families and research depth.
3. Search every applicable family A-X.
4. Missing metrics are explicitly PARTIAL, UNAVAILABLE, or EXCLUDED.
5. Do not count correlated metrics as independent evidence merely because they have different names.
6. Do not treat reconstructed metrics as direct observations.
7. Do not reverse-engineer hidden General/Specialist internals.
8. Do not use private physiological or instrumented tracking information unless actually published.
9. Preserve source URLs/citations for public evidence in the completed audit.
10. Final color is prohibited until the Master Metrics Sweep and the existing Completion Ledger are both satisfied.

## Y. ADVANCED PUBLIC / RECONSTRUCTABLE DECIDING SIGNALS

Coverage rule: Run every item below when the necessary public inputs exist. Charting-derived fields must include charted-match sample size and coverage warnings. Derived fields must expose their inputs and be labeled RECONSTRUCTED. If inputs are insufficient, mark PARTIAL or UNAVAILABLE.

### Y1. Serve +1 / Return Neutralization

- Serve +1 conversion / <=3-shot service-point win%.
- Serve Impact where publicly charted.
- Return-in-play%.
- Return-in-play win%.

Tennis Matrix AI - Reusable MAX Metrics Plus Audit

Page 17

- Return neutralization rate.
- Return winner%.
- Return Aggression Score where charted.
- Deep Return Rate / Return Depth Index.
- First-serve return depth.
- Second-serve return depth.
- Return slice/chip frequency.
- Return-depth differential.
- Serve +1 vs opponent return-neutralization differential.

### Y2. Rally-Length Matchup

- Average rally length.
- Rally length on service points.
- Rally length on return points.
- 1-3 shot win%.
- 4-6 shot win%.
- 7-9 shot win%.
- 10+ shot win%.
- Rally-length distribution by bucket.
- Rally-length matchup differential.
- Likely-rally-distribution advantage.
- Rally-Length Control Score.
- Short-rally advantage.
- Medium-rally advantage.
- Long-rally advantage.
- Rally Aggression Score where charted.

### Y3. Wing / Direction / Tactical Shot Patterns

- Forehand potency per match and per 100 forehands.
- Backhand potency per match and per 100 backhands.
- Forehand share of groundstrokes / forehand dependency.
- Backhand slice frequency.
- Forehand winner%.
- Forehand down-the-line winner%.
- Forehand inside-out winner%.
- Backhand winner%.

- Backhand down-the-line winner%.
- Serve-and-volley frequency.
- Serve-and-volley win%.
- Net frequency.
- Net win%.
- Net effectiveness.
- Drop-shot frequency.
- Drop-shot winner/effectiveness%.

Tennis Matrix AI - Reusable MAX Metrics Plus Audit

Page 18

- Serve direction wide/body/T when charted.
- Serve-direction predictability/concentration.
- Serve-location matchup matrix.
- Return direction/depth patterns.
- Crosscourt/down-the-line/inside-out/inside-in tendencies when charted.
- Directional Vulnerability Score.
- Weak-Wing Targetability.
- Pattern Exploitability Score.
- Plan-B Score.

All tactical scores not directly published must be transparently reconstructed from charted inputs.

#### Y4. Serve / Return Dependency & Exposure

- Serve Dependency Score.
- Return Dependency Score.
- First-Serve Miss Penalty.
- Second-Serve Exposure Rate.
- Second-Serve Punishment Differential.
- Break-Point Creation Quality.
- Break Opportunity -> Break Conversion Pipeline.
- Hold Under Pressure Differential.
- Break Consolidation Differential.
- Rebreak Vulnerability.
- Immediate rebreak vulnerability.
- Serve quality when first-serve percentage falls below player baseline.
- Return quality when opponent first serve exceeds baseline.

#### Y5. Score-State / Closing Dynamics

When point/game logs support reconstruction:

- Closing Degradation Index.
- Lead Conversion Curve.
- Deficit Recovery Curve.
- One-break-up conversion.
- One-set-up conversion.
- Set-plus-break-up conversion.
- One-break-down recovery.
- One-set-down recovery.
- Set-plus-break-down recovery.

- Score-state performance at 0-0, 15-15, 30-30, deuce and break point.
- Pressure Leverage Performance using transparent win-probability leverage.
- Pressure Regression Score.
- Tiebreak Entry Quality.
- Decider Entry Quality.
- Late-set hold/break differential.
- 4-4 / 5-5 performance when sample permits.

Tennis Matrix AI - Reusable MAX Metrics Plus Audit

Page 19

#### Y6. Match-Control / Scoreline Quality

- Match Control Index.
- Scoreline Deception Index.
- Dominant-Loss detection.
- Weak-Win detection.
- Underlying point dominance in wins/losses.
- Performance vs Expected Opponent Strength.
- Actual-minus-expected point win%.
- Actual-minus-expected hold%.
- Actual-minus-expected break%.

These must be reconstructed with stated baselines and sample windows.

#### Y7. Opponent-Adjusted Skill Residuals

- Opponent-adjusted point win%.
- Opponent-adjusted hold%.
- Opponent-adjusted break%.
- Opponent-adjusted first-serve points won%.
- Opponent-adjusted second-serve points won%.
- Opponent-adjusted first-serve return points won%.
- Opponent-adjusted second-serve return points won%.
- Opponent-adjusted return points won%.
- Expected-vs-Actual Serve Performance.
- Expected-vs-Actual Return Performance.
- Residual Serve Form.
- Residual Return Form.
- Skill Momentum - serve.
- Skill Momentum - return.
- Serve/Return Divergence.
- Surface x recency x opponent-adjusted serve rating.
- Surface x recency x opponent-adjusted return rating.

#### Y8. Distribution / Volatility / Floor-Ceiling

- Rolling match-level performance volatility.
- Serve volatility.
- Return volatility.
- Point-win volatility.
- Hold/break volatility.
- Consistency Score.



- Performance floor: 10th/20th percentile.
- Performance ceiling: 80th/90th percentile.
- Bad-Day Vulnerability.
- Upside Frequency.
- Favorite Floor vs Underdog Ceiling Overlap.

Tennis Matrix AI - Reusable MAX Metrics Plus Audit

Page 20

- Underdog Floor vs Favorite Ceiling Overlap.
- Full Performance Distribution Overlap.
- Matchup Uncertainty Score.
- Small-sample uncertainty expansion.
- Lower-tour uncertainty expansion.
- Surface-transition uncertainty expansion.
- Missing-data uncertainty expansion.

Y9. Shrinkage / Sample Stabilization

When enough historical data exists:

- Bayesian/shrinkage serve strength.
- Bayesian/shrinkage return strength.
- Shrunken hold%.
- Shrunken break%.
- Shrunken first-serve points won%.
- Shrunken second-serve points won%.
- Shrunken return-points-won%.
- Surface-specific partial pooling.
- Recency-weighted partial pooling.
- Tour-level prior adjustment.
- Effective sample-size adjustment.

Always show the prior/baseline and method; label as RECONSTRUCTED OUTSIDE ESTIMATE.

Y10. Opponent Archetype / Similarity

- Objectively defined opponent archetype.
- Performance vs big servers.
- Performance vs elite returners.
- Performance vs aggressive baseliners.
- Performance vs defensive/counterpunch profiles.
- Performance vs short-rally specialists.
- Performance vs long-rally specialists.
- Archetype-adjusted win rate.
- Archetype-adjusted point win%.
- Transparent Archetype Elo/strength estimate.
- Similarity-Weighted Opponent Performance.
- Similar-opponent surface performance.
- Similar-opponent serve/return performance.
- Handedness-adjusted serve/return splits.
- Lefty serve-pattern exposure.
- One-handed vs two-handed backhand split where sufficient sample exists.

#### Y11. Rematch Learning / Tactical Adaptation

- Rematch Learning Rate.
- Performance change across repeated meetings.

Tennis Matrix AI - Reusable MAX Metrics Plus Audit

Page 21

- Serve-placement change between meetings.
- Return-position change.
- Rally-length change.
- Net-frequency change.
- Second-serve attack change.
- Tactical Adaptation Score.
- Prior-loss correction evidence.
- Prior-win repeatability evidence.

#### Y12. Current-Tournament Form

- Round-by-round performance progression.
- Current-Tournament Serve Form.
- Current-Tournament Return Form.
- Current-Tournament Point Dominance.
- Current-Tournament Hold/Break Differential.
- Current-Tournament Opponent-Adjusted Dominance.
- Current-Tournament Tiebreak/pressure results.
- Current-Tournament Set-1 performance.
- Current-Tournament closing performance.
- Current-Tournament workload efficiency.
- Efficiency of Wins.
- Cost of Advancement.

#### Y13. Schedule / Adaptation / Travel Proxies

- Rest-Adjusted Workload Differential.
- Schedule Compression Index.
- Travel Burden Proxy.
- Time zones crossed.
- Days between previous and current tournament.
- Surface Adaptation Time.
- Matches already played on current surface.
- Qualifier Adaptation Effect.
- Qualifier Workload Cost.
- Bye/Rust Effect.
- Walkover rhythm/rest effect.
- Consecutive-tournament burden.

Never convert these proxies into private physiological claims.

#### Y14. Environmental Sensitivity

When historical weather/venue data can be joined reliably:

- Weather Serve Sensitivity.
- Weather Return Sensitivity.
- Heat performance split.

- Humidity performance split.

Tennis Matrix AI - Reusable MAX Metrics Plus Audit

Page 22

- Wind Sensitivity Proxy.
- Ace change in wind.
- Double-fault change in wind.
- First-serve-in change in wind.
- Altitude Serve Boost Sensitivity.
- Altitude return penalty/benefit.
- Court-Speed Sensitivity.
- Fast-court serve uplift.
- Slow-court return uplift.
- Day/night split.
- Best-of-3 vs Best-of-5 translation when applicable.

Require meaningful sample size; otherwise mark suggestive/partial.

#### Y15. ADVANCED MATCHUP CROSS-FEATURES

Do not evaluate the advanced metrics only in isolation. Build the following opponent-specific cross-features when inputs exist:

- P1 Serve +1 vs P2 Return Neutralization.
- P2 Serve +1 vs P1 Return Neutralization.
- P1 second-serve exposure x P2 second-serve punishment.
- P2 second-serve exposure x P1 second-serve punishment.
- P1 short-rally win% vs P2 short-rally defense.
- P1 long-rally win% vs P2 long-rally tolerance.
- Reverse all rally interactions for P2.
- Serve-direction concentration vs opponent return-direction strength.
- Weak-wing targetability vs opponent directional preference.
- Net frequency/success vs opponent passing effectiveness.
- Drop-shot use vs opponent forward-movement vulnerability when publicly supported.
- Favorite closing degradation vs underdog deficit recovery.
- Favorite rebreak vulnerability vs underdog break-back strength.
- Favorite floor vs underdog ceiling.
- Favorite volatility vs underdog upside frequency.
- Favorite current-tournament serve form vs underdog current-tournament return form.
- Favorite current-tournament return form vs underdog current-tournament serve form.
- Favorite schedule compression vs underdog rest/adaptation advantage.
- Similarity-weighted underdog performance vs favorite archetype.
- Similarity-weighted favorite performance vs underdog archetype.

#### Y16. ADVANCED METRICS RELIABILITY GATE

For every Y-family signal record:

- Public source(s).
- Direct vs reconstructed.
- Date/window.
- Surface.

Tennis Matrix AI - Reusable MAX Metrics Plus Audit

Page 23

- Number of matches.
- Number of charted matches if applicable.
- Number of points/service games/return games if known.
- Opponent-strength adjustment status.
- Coverage bias risk.
- Reliability: HIGH / MEDIUM / LOW.
- Decision weight: FULL / REDUCED / CONTEXT ONLY / ZERO.
- Whether the signal is independent or correlated with another metric.

Charting warning: Match Charting Project samples can represent only a fraction of a player's matches and may be surface/opponent-skewed. Never give a charting-derived signal full weight without reporting coverage.

#### Y17. FALSE-FAVORITE / TRAP RANGE TEST

Before a favorite can receive DOUBLE GREEN or GREEN, explicitly test:

1. Does the favorite's realistic performance floor overlap the underdog's realistic ceiling?
2. Does the underdog own at least one high-reliability opponent-specific cross-feature?
3. Is the favorite's recent record stronger than its underlying point/hold/break performance?
4. Is the favorite benefiting from unsustainable tiebreak/break-point/deciding-set overperformance?
5. Does opponent adjustment materially shrink the favorite's apparent edge?
6. Does current-tournament form contradict the longer-term rating?
7. Does workload/schedule compression materially narrow the matchup?
8. Does archetype/similarity evidence favor the underdog?
9. Do charted tactical patterns reveal a direct exploitable collision?
10. Does distribution overlap make the headline probability look too certain?

Report:

- Floor/Ceiling Overlap: LOW / MODERATE / HIGH.
- Distribution Overlap: LOW / MODERATE / HIGH.
- Opponent-Specific Cross-Feature Risk: LOW / MODERATE / HIGH.
- Regression Risk: LOW / MODERATE / HIGH.
- Advanced Trap Evidence Count: [N], counting only genuinely independent reliable signals.

#### REUSABLE PER-MATCH EXECUTION TEMPLATE

Match Identification

- Player 1: [PLAYER 1]
- Player 2: [PLAYER 2]
- Tournament: [TOURNAMENT]
- Level: [LEVEL]
- Surface: [SURFACE]
- Round: [ROUND]
- Matrix predicted winner: [PLAYER]
- Headline win probability: [WP]
- Recommendation: [LABEL]

Tennis Matrix AI - Reusable MAX Metrics Plus Audit

Page 24

- Upset Risk: [LEVEL]
- Data Quality: [DQ]
- Model Agreement: [LABEL]
- Matchup Closeness: [LABEL]

## 1. Matrix Component Verification

For every supplied live component, record:

- Surface Elo: favored player, raw probability, effective weight/contribution, reliability, availability, sample, status.
- Serve & Return: favored player, raw probability, effective weight/contribution, reliability, availability, sample, status.
- Recent Form: favored player, raw probability, effective weight/contribution, reliability, availability, sample, status.
- H2H: favored player, raw probability, reliability, sample, status; exclude from decision weight when the Matrix excludes it.
- Market Consensus: favored player, raw probability, effective weight/contribution, reliability, availability, sample, status.
- General Model: report the supplied output only. Do not invent or reverse-engineer hidden internal features, weights, training logic, calibration, or exact evidence overlap.
- Specialist Model: report the supplied output only. Do not invent or reverse-engineer hidden internal features, weights, training logic, calibration, or exact evidence overlap.
- Monte Carlo: favorite support, simulation range, straight-set probabilities, and headline-vs-MC gap when supplied.

## 2. Player 1 Public Verification

Research and classify each item as STRONG, WEAK, CONFLICT, UNAVAILABLE, or EXCLUDED:

- Current ranking and ranking trajectory.
- Last 5 and last 10 results.
- Current streak.
- Same-surface recent record.
- Current tournament results.
- Quality wins and bad losses.
- Opponent quality.
- Public serve/return statistics where available.
- Set 1 record and early-start pattern where reconstructable.
- Deciding-set and tiebreak record where reconstructable.
- Break-point and closing evidence where reconstructable.
- Common-opponent performance.
- H2H and same-surface H2H.
- Recent workload: matches, sets, games, minutes, consecutive-day play, qualifying workload, late finishes.
- Public injury, illness, withdrawal, retirement, medical-timeout, or return-from-injury information.
- Publicly documented travel/rest context.
- Public tactical/style evidence: handedness, serve profile, return profile, rally tolerance, surface specialization, and documented matchup tendencies.
- Current market and line movement where available.

## 3. Player 2 Public Verification

Run the exact same public verification fields and evidence classifications used for Player 1. Do not give the Matrix favorite more research depth than the opponent.

## 4. Direct Player 1 vs Player 2 Differential

Compare both players directly on:

Tennis Matrix AI - Reusable MAX Metrics Plus Audit

Page 25

- Surface strength.
- Current form.
- Opponent-adjusted recent performance.
- Serve.
- Return.
- Second-serve vulnerability/attack.

- Set 1 performance.
- Deciding-set performance.
- Tiebreak performance.
- Closing/lead durability.
- Common opponents.
- H2H/rematch evidence.
- Workload/rest.
- Injury/availability.
- Ranking trajectory/development.
- Style/tactical matchup.
- Current market.
- Monte Carlo and expected match length.

#### 5. Favorite Loss Autopsy

Using recent publicly documented losses:

- Opponent and opponent quality.
- Surface and tournament level.
- Score.
- Set 1 result.
- Lead/collapse evidence.
- Serve and return performance where published.
- First-/second-serve issues where published.
- Break-point and break-consolidation evidence.
- Tiebreak/deciding-set evidence.
- Short-rally/long-rally evidence where publicly documented.
- Physical/injury evidence only when publicly reported.
- Repeatable tactical failure modes.

Report 3-5 documented favorite failure modes when evidence supports them.

#### 6. Predicted-Loser Win Autopsy

Using recent publicly documented wins:

- Opponent and opponent quality.
- Surface and tournament level.
- Score.
- Whether opponent was favored/higher ranked.
- Set 1 recovery or lead protection.
- Serve dominance.
- Return pressure.

Tennis Matrix AI - Reusable MAX Metrics Plus Audit

Page 26

- Second-serve attack.
- Break-point performance.
- Comeback/closing.
- Tiebreak performance.
- Rally/pace tolerance where documented.
- Repeatable tactical pathway.

Separate repeatable pathways from one-off variance.

## 7. Opponent-Specific Interaction Reconstruction

When public statistics support it, reconstruct or estimate transparently:

- Expected hold differential.
- Expected break differential.
- Second-serve attack mismatch.
- Serve-vs-return matchup.
- Surface-adjusted form differential.
- Common-opponent differential.
- Workload/rest differential.

Any reconstructed metric must show its public inputs and method. Do not label an outside approximation as a proprietary Tennis Matrix metric.

## 8. Favorite Failure-Mode Compatibility

For each documented favorite failure mode, ask whether this opponent can create it:

- YES - strong evidence.
- POSSIBLY - partial evidence.
- NO - opponent lacks the weapon.
- UNAVAILABLE - evidence cannot be reconstructed.

Report the compatibility count.

## 9. Underdog Win Pathways

Assess all 15 reusable pathways:

1. Serve-through.
2. Return-pressure.
3. Second-serve exploitation.
4. Short-rally.
5. Long-rally.
6. Movement/physical using only public observable/reportable proxies.
7. Slow-start/Set 1.
8. Deciding-set.
9. Tiebreak.
10. Workload/fatigue proxy.
11. Style mismatch.
12. Market-information.
13. Favorite-collapse/closing.
14. Surface-transition.
15. Recent-improvement/ranking-lag.

Tennis Matrix AI - Reusable MAX Metrics Plus Audit

Page 27

Label each REALISTIC, WEAK, NOT SUPPORTED, or UNAVAILABLE. Count only genuinely independent REALISTIC pathways.

## 10. Evidence Independence

Do not claim exact hidden General/Specialist circularity. Instead:

- Count raw supplied component agreement.
- Separate clearly independent public evidence families.
- Flag possible/unknown model overlap as UNKNOWN when internal composition is not supplied.
- Treat market as a separate evidence family when sourced independently.
- Treat current injury/news as a separate family when publicly documented.

Report Evidence Independence Risk: LOW / MODERATE / HIGH / UNKNOWN.

### 11. Public/Reconstructable Stress Test

Do not claim to recompute the proprietary Tennis Matrix engine unless its real blending/calibration formula is supplied.

Instead test transparent outside scenarios:

- Reduce the observed Elo edge.
- Reduce recent-form edge.
- Reduce serve edge.
- Improve opponent return assumptions.
- Worsen public workload/rest assumptions.
- Move market probability toward the opponent.
- Remove one supplied component from the qualitative evidence count.

Report whether the external evidence case remains robust, becomes contested, or becomes fragile. Label this an OUTSIDE STRESS TEST, not an exact Matrix winner-switch threshold.

### 12. Public Historical Analogs

Search public/reconstructable match history for comparable matches using available dimensions such as:

- Ranking/Elo gap.
- Surface.
- Recent-form gap.
- Serve/return profile.
- Market probability.
- Workload.
- Style/archetype.
- Tournament level.

Report comparable examples and explain similarity. Do not claim they are the private 100-match workbook or use invented proprietary similarity scores.

### 13. Fresh External Validation

Freshly verify both players:

- Exact matchup/current draw validity.
- Tournament, level, surface, round.
- Withdrawal/replacement/walkover status.
- Rankings.
- Last 5/10.

Tennis Matrix AI - Reusable MAX Metrics Plus Audit

Page 28

- Same-surface form.
- H2H.
- Injuries/availability.
- Current tournament workload.
- Current market from multiple sources where available.
- Current draw/status.

Classify external evidence as CONFIRMS FAVORITE, WEAKLY CONFIRMS FAVORITE, NEUTRAL, WEAKLY CONFIRMS UNDERDOG, or CONFIRMS UNDERDOG.

### 14. Contradiction Ledger

List every meaningful contradiction:

- Matrix headline vs Surface Elo.
- Matrix headline vs Serve/Return.



- Matrix headline vs Recent Form.
- Matrix headline vs Market.
- Matrix headline vs General output.
- Matrix headline vs Specialist output.
- Matrix headline vs Monte Carlo.
- Matrix headline vs current public form.
- Matrix headline vs same-surface evidence.
- Matrix headline vs injury/workload evidence.
- Matrix headline vs current market.

Do not count unavailable or excluded evidence as disagreement.

#### 15. Final Verification Classification

Assign only after all available layers are completed:

- DOUBLE GREEN - strongest alignment between Matrix evidence and independent public verification, with no material unresolved contradiction.
- GREEN - favorable overall, with limited non-critical contradictions.
- YELLOW - meaningful conflict, thin evidence, or opponent has credible independent win pathways.
- RED - material reliable evidence undermines the Matrix pick or the favorite has multiple opponent-specific failure conditions.
- PASS / INCOMPLETE - required current information cannot be verified sufficiently.

Report:

- Evidence winner.
- Strongest 3 reasons.
- Strongest contradiction.
- Independent realistic underdog pathways count.
- Public-source confidence.
- Final color/action.
- Missing or partial fields.

#### COMPLETION LEDGER - EVERY MATCH

Before calling a match complete, verify:

- Matrix favorite evaluated separately.

Tennis Matrix AI - Reusable MAX Metrics Plus Audit

Page 29

- Opponent evaluated separately.
- Every supplied Matrix component checked for reliability/sample/status.
- Fresh public research completed for both players.
- Direct differential completed.
- Favorite loss autopsy completed or marked UNAVAILABLE.
- Predicted-loser win autopsy completed or marked UNAVAILABLE.
- Common-opponent layer completed or marked UNAVAILABLE.
- Set 1 / deciding-set / tiebreak layer completed or marked UNAVAILABLE.
- Workload/rest layer completed or marked PARTIAL/UNAVAILABLE.
- Injury/availability layer completed or marked UNAVAILABLE.
- Favorite failure modes identified.
- Opponent-specific compatibility assessed.
- All 15 win pathways classified.
- Evidence-independence review completed without inventing hidden model internals.
- Outside stress test completed without claiming proprietary recomputation.

- Public historical analog search completed or marked UNAVAILABLE.
- Current market checked from multiple sources where available.
- Current draw/withdrawal validity checked.
- Contradiction ledger completed.
- Advanced Y1-Y14 metric families screened for both players.
- Advanced Y15 opponent-specific cross-features built where inputs exist.
- Y16 advanced-metrics reliability gate completed.
- Y17 false-favorite / trap range test completed.
- Charting coverage/sample bias explicitly checked where charting data was used.
- Floor/ceiling and performance-distribution overlap assessed where enough data exists.
- Final classification assigned only after the above.

#### Hard Stop Rule

If a mandatory layer has not been run and is not legitimately UNAVAILABLE/PARTIAL, report:

INCOMPLETE AUDIT - missing: [specific layer/source].

Never claim private physiological measurements, instrumented tracking data, hidden model internals, the private independent Green list, or the private 100-match workbook unless those sources are actually supplied.

#### ADDED VALIDATION RULES - CURRENT-STRENGTH & FALSE-GREEN

##### CONTROL

These rules are mandatory additions to the Full Verification Audit. They use only public or transparently reconstructable evidence and are designed to prevent long-term reputation/rating strength from masking present-level uncertainty.

##### Current-Strength / Ranking-Lag Gate

Calculate current-strength separately from long-term ranking/Elo when recent opponent-adjusted performance materially exceeds the player's trailing baseline. Include ranking velocity, Elo velocity where available, serve/return trajectory, quality-of-opposition progression, and recent surface performance.

##### Recent Elite-Win Escalation

A recent Top-10/20/50 quality win triggers a repeatability check: underlying points/games, serve/return performance, opponent condition, surface, and whether the performance repeated across multiple matches.

##### Breakthrough-Player Detection

Flag players whose last 8-12 weeks materially outperform their trailing 12-month baseline. Reweight recent opponent-adjusted evidence upward only when supported by adequate sample and underlying performance.

##### Trajectory Differential

Compare both players' ranking velocity, Elo/strength velocity, opponent-adjusted form velocity, serve trajectory, return trajectory, quality-of-opposition progression, and current-surface trajectory.

##### Current-Tournament Evidence Gate

Compare current-tournament serve, return, point dominance, hold/break, opponent quality, workload, conditions, and round-by-round progression.

A player with no current-condition matches receives added uncertainty rather than assumed historical form.

##### Long-Layoff / Rust Uncertainty

A meaningful competitive layoff widens the player's performance distribution. Historical strength may remain relevant but cannot erase uncertainty about present floor, current conditions, and match sharpness.

##### Rust x Momentum Interaction

Explicitly test established/returning favorite versus match-sharp or rapidly improving opponent. Treat the interaction as a cross-feature, not two unrelated notes.

##### Information-Asymmetry Penalty

When one player's present level is well observed and the other's evidence is mainly historical, reduce confidence in the less-observed player's current estimate.

## Probability Compression Under Uncertainty

When present-level uncertainty is high, widen the interval and compress extreme reconstructed probabilities toward a more conservative range.

State the reason and method.

## Floor / Ceiling Revaluation

Re-estimate the favorite's realistic lower tail after layoff/inactivity and the underdog's upper tail after recent elite-quality performances. Re-run distribution overlap.

## Age x Layoff Interaction

Age alone is not a removal signal. Age combined with meaningful inactivity, absent current-condition evidence, and declining/recently volatile performance increases uncertainty and widens the floor distribution.

## Legacy / Name Bias Check

Remove player names and career accomplishments conceptually. Ask whether last 10-20 matches, surface evidence, opponent quality, trajectory, workload, and current-condition evidence still support the same verification color.

## Market-Veto Investigation

A major gap between Matrix/independent strength and a liquid market temporarily blocks Green until the discrepancy is investigated. Market is not automatically correct, but unexplained disagreement is a mandatory contradiction.

## False-Green Feedback Rule

Every Green/Double-Green loss receives a pre-match failure review. Record which public signals existed before first serve, which gates failed to trigger, and whether the pattern should downgrade future structurally similar matches.

## Pre-Match Removal Question

Before Green, answer: What is the strongest evidence that this favorite should be removed? Green requires an evidence-based explanation for why that removal case is insufficient.

## INTEGRATED FINAL DECISION ARCHITECTURE

Mandatory addition to the Verification Audit. This section controls how Verification, Disagreement, Dangerous Underdog, stress testing, and final list eligibility interact.

### 1. Final Combined Decision Gate

A candidate reaches the Final Green List only if the Verification Audit approves it AND the Disagreement/Trap Audit fails to falsify it AND the Dangerous Underdog Screen is clear AND the final stress/removal tests are passed. Failure at any required stage blocks Green.

### 2. Disagreement Veto Power

A RED/PASS, unresolved CRITICAL contradiction, or active Dangerous Underdog GREEN LOCK from the Disagreement Audit automatically prevents Green regardless of the Matrix probability or Verification approval.

### 3. Hard Downgrade Triggers

Specified combinations must mechanically downgrade rather than merely warn. Example: rapid-riser/ranking-lag opponent + credible recent elite performance + favorite inactivity/current-level uncertainty locks Green until escalation is passed.

### 4. Minimum Independent-Evidence Requirement

Do not approve from raw metric count. Green requires at least 4 genuinely independent reliable evidence families supporting the candidate after contradictions; Double Green requires at least 6. Correlated metrics collapse into one effective family.

### 5. Contradiction Budget

Double Green: zero unresolved CRITICAL contradictions and no more than one unresolved MAJOR contradiction. Green: zero unresolved CRITICAL contradictions and only limited MAJOR contradictions that are explicitly defeated. Otherwise downgrade.

### 6. Evidence Freshness Score

Report how much decisive evidence comes from the last 4/8/12/26 weeks and how much is older. Stale evidence cannot carry the same confidence as fresh, surface-relevant, opponent-adjusted evidence.

### 7. Historical Strength vs Current Strength Split

Finish every match with separate Historical Strength and Current Estimated Strength assessments. A material divergence is a mandatory warning and triggers deeper current-level review.

#### 8. Probability Stress Range

Report Base, Conservative, and Optimistic reconstructed probability/range. If reasonable conservative assumptions push the favorite near 50% or flip the winner, Green is fragile and must be downgraded unless independently resolved.

#### 9. Removal-First Final Stage

Before approval, construct the strongest evidence-based REMOVE CASE against the Matrix winner. The candidate must defeat that case using independent reliable evidence. Failure to defeat it blocks Green.

#### 10. False-Green Learning Ledger

Every Green/Double-Green loss receives a pre-match failure classification using only information available before first serve: rapid riser missed, stale rating, weak-win inflation, current decline, matchup collision, market contradiction, inactivity, etc. Recurrent structures become future downgrade triggers.

#### 11. Calibration by Audit Color

Track actual outcomes separately for Double Green, Green, Yellow, Red, and PASS/removed candidates. If higher colors do not materially outperform lower colors over adequate samples, tighten thresholds.

#### 12. Audit Confidence Is Not Win Probability

Always report Audit Confidence separately from estimated win probability. A high estimated probability with incomplete/low-quality evidence cannot receive equally high audit confidence.

#### 13. Minimum Coverage Threshold

Report Audit Coverage %. Double Green requires very high coverage of CRITICAL evidence families; Green requires sufficient critical coverage. Missing supplemental metrics alone do not block approval.

#### 14. Critical-Metric Coverage

Classify audit fields as CRITICAL / IMPORTANT / SUPPLEMENTAL. Missing current surface form, serve/return, opponent quality, availability, or present-strength evidence matters far more than missing an obscure tactical statistic.

#### 15. Effective Independent Evidence Count

Final output must report the number of effective independent evidence families after correlated-data compression. Do not allow dozens of correlated metrics to create false unanimity.

#### 16. Mandatory Strongest-Underdog Table

For every proposed Green, list the underdog's 5 strongest pre-match arguments, reliability/sample, evidence family, and whether each was DEFEATED / UNRESOLVED / SUPPORTS REMOVAL.

#### 17. Final Rejection Reason

Every removed/downgraded favorite receives one concise human-readable reason identifying the decisive structural risk rather than merely listing metrics.

#### FINAL INTEGRATED WORKFLOW

MANDATORY WORKFLOW: Tennis Matrix Candidate -> Verification Audit -> Disagreement/Trap Audit -> Dangerous Underdog Screen -> Stress Test -> Removal Case -> FINAL GREEN LIST

No candidate may bypass a stage. A blocked stage must be resolved or the candidate remains downgraded/removed. The purpose is filtering, not forcing a winner.